

Unequal Cost Load-Balancing

Let's configure variance and feasible successors to achieve unequal cost load balancing. On R1, let's make the R1 to R2 more expensive:

```
conf t
interface GigabitEthernet0/0
  delay 100
end
```

Now, let's introduce variance:

```
conf t
router eigrp ENTERPRISE
  address-family ipv4 unicast autonomous-system 100
    topology base
      variance 128
    exit-af-topology
  exit-address-family
end
```

Let's check the route to 5.5.5.5/32 to see multiple paths available:

```
show ip eigrp topology 5.5.5.5/32
```

```

R1#show ip eigrp topology 5.5.5.5
EIGRP-IPv4 VR(ENTERPRISE) Topology Entry for AS(100)/ID(1.1.1.1)
Entry 5.5.5.5/8 not in topology table
R1#show ip eigrp topology 5.5.5.5/32
EIGRP-IPv4 VR(ENTERPRISE) Topology Entry for AS(100)/ID(1.1.1.1) for 5.5.5.5/32
State is Passive, Query origin flag is 1, 2 Successor(s), FD is 2703360, RIB is 21120
Descriptor Blocks:
10.0.12.2 (GigabitEthernet0/0), from 10.0.12.2, Send flag is 0x0
Composite metric is (66928640/1392640), route is Internal
Vector metric:
  Minimum bandwidth is 1000000 Kbit
  Total delay is 1011250000 picoseconds
  Reliability is 255/255
  Load is 1/255
  Minimum MTU is 1500
  Hop count is 2
  Originating router is 5.5.5.5
10.0.14.2 (GigabitEthernet0/1), from 10.0.14.2, Send flag is 0x0
Composite metric is (2703360/2048000), route is Internal
Vector metric:
  Minimum bandwidth is 1000000 Kbit
  Total delay is 31250000 picoseconds
  Reliability is 255/255
  Load is 1/255
  Minimum MTU is 1500
  Hop count is 3
  Originating router is 5.5.5.5

```

sh ip route 5.5.5.5

```

R1#sh ip route 5.5.5.5
Routing entry for 5.5.5.5/32
  Known via "eigrp 100", distance 90, metric 21120, type internal
  Redistributing via eigrp 100
  Last update from 10.0.12.2 on GigabitEthernet0/0, 00:19:36 ago
  Routing Descriptor Blocks:
    * 10.0.14.2, from 10.0.14.2, 00:19:36 ago, via GigabitEthernet0/1
      Route metric is 21120, traffic share count is 24
      Total delay is 32 microseconds, minimum bandwidth is 1000000 Kbit
      Reliability 255/255, minimum MTU 1500 bytes
      Loading 1/255, Hops 3
    10.0.12.2, from 10.0.12.2, 00:19:36 ago, via GigabitEthernet0/0
      Route metric is 522880, traffic share count is 1
      Total delay is 1012 microseconds, minimum bandwidth is 1000000 Kbit
      Reliability 255/255, minimum MTU 1500 bytes
      Loading 1/255, Hops 2
R1#

```